# ARPIT K. JAISWAL

**GAMEPLAY & AI SYSTEMS DEVELOPER** [ UNITY DEVELOPER ] [ 3.5+ YRS EXPERIENCE ]

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## EXECUTIVE PROFILE

Core Specializations & Studio Focus

- **Production & Multiplayer Systems (Unity C#):** Gameplay & Systems Engineer with 3.5+ years shipping 3 Google Play titles and real-time multiplayer games using Photon (PUN 2), Stockfish AI, and deterministic FSMs.
- **Architecture, Native Bridges & 60 FPS:** Specialist in procedural N×N mesh/puzzle generation, native Android (JNI) / iOS (Obj-C) bridges, non-blocking OTA content delivery, and Zero-GC object pooling locked at 60 FPS.
- **AI-Augmented Tooling & LiveOps:** Leverages Claude, ChatGPT/Codex, and custom Unity EditorWindows to accelerate URP/HLSL shader prototyping, offline localization pipelines, and Google/Apple IAP monetization.

## PRODUCTION TRACK RECORD (KPIS)

Quick 6-Second Recruiter Highlights

### 3.5+ Years — STUDIO EXPERIENCE

- **Vasundhara Game Studio, Surat (2yr)** • **Dec 2024 – Jan 2025 & Dec 2025 – Present**
- **StudioKrew, Delhi (3m)** • Mar 2024 – May 2024
- **Appzia, Pune (1yr)** • June 2024 – July 2025

### 3 Shipped Titles — GOOGLE PLAY

- **Calm Jigsaw** [Store ↗]
- **MeowSweeper** [Store ↗]
- **Calm Solitaire** [Store ↗]

### Engineered Subsystems [Unity Playable Demo ↗]

- **Procedural Jigsaw**
- **Jigsaw Shader Mask**
- **Native Image Bridge**
- **Level Sync (OTA)**
- **Live-Ops Backend**
- **Offline Leaderboard**
- **IAP Currency State**
- **Argos Localization**
- **Asset Organizer**
- **Arrow Strike**

### Early-Career Prototypes [GitLab Repo ↗] [Demos ↗]

This Is Blast • Arrow Strike • Tank Battle • Bullet Chess • Tycoon RTS • Trivologist

## PROFESSIONAL STUDIO EXPERIENCE

Commercial Production Roles & Systems

### Vasundhara Game Studio | AI Product Developer — Unity Systems & Gameplay [ CURRENT ]

Surat, GJ • **Dec 2024 – Jan 2025 & Dec 2025 – Present**

- **Procedural Jigsaw & Shader Mask (Calm Jigsaw):** Architected N×N Bezier puzzle generator using 9 mathematical edge signatures, dynamic UV quad meshes, and custom URP/HLSL shader masking with zero pre-sliced art.
- **Native Image Bridge & OTA Sync:** Engineered C# bridge over native Android (JNI) / iOS (Obj-C) APIs for 3-tier RAM LRU and disk caching, texture downsampling, and non-blocking OTA manifest level delivery.
- **Hybrid Logic & Hint-Cascade Solver (MeowSweeper):** Built hybrid Sudoku/Minesweeper/Star-Battle puzzle engine with backtracking hint-cascade solver FSM, discrete multi-touch locks, and IAP powerup wallet.
- **Card Seam Merging & Arrow Mechanics (Calm Solitaire & Arrow):** Engineered rectangular card-grid 0px seam merging, displacement swapping, procedural flame noise shaders, and outside-in peeling arrow escape mechanics.
- **Offline Leaderboard, IAP & Editor Tooling:** Built zero-server bot-paced seasonal leaderboards with anti-rollback clamping, Google/Apple IAP state machines, Asset Organizer EditorWindow, and Argos IPC localization.

## PROFESSIONAL STUDIO EXPERIENCE (CONTINUED)

Multiplayer, AI & Standalone Game Architecture

### Appzia | Unity3D Systems & Gameplay Developer

**Pune, India • June 2024 – July 2025**

- **Stockfish AI & Real-Time Chess (Bullet Chess):** Architected multiplayer chess engine using Photon (PUN 2) and Stockfish API for live move evaluation, client-side board validation, clock sync, and tournament brackets.
- **Arcade Simulation & Economy Systems (Pinnacle):** Led development of Pinnacle arcade simulation replicating physical machines, implementing deterministic FSMs, secure Apple/Google IAP, and Zero-GC object pooling at 60 FPS.

### StudioKrew | Unity Game Developer (Remote)

**Delhi, India (Remote) • March 2024 – May 2024**

- **Solo Card Game Architecture (FreeCell & Solitaire):** Delivered complete end-to-end development independently, architecting deterministic card stack rules, memento-based undo/redo state history, and responsive touch controls.
- **Dice Physics & Scoring State Machine (Yahtzee):** Engineered probabilistic dice-rolling physics, category score validation state machines, and turn-based round flow with fluid particle animations.
- **Grid Puzzle Engine & Serialization (BrokenTiles):** Developed reactive grid-matching algorithms with cascading tile clearing, implementing lightweight JSON save serialization for instant session recovery.

## TECHNICAL SKILLS & ENGINE STACK

Production Engineering & Tooling Matrix

**Revenue Systems & Product Growth**

Google / Apple IAP • Ads Mediation • Event Funnels & Attribution • User Engagement & Session Loops • Revenue Tracking & Cohort Analytics

**Rendering & Optimization**

URP • SRP Batcher • GPU Instancing • ASTC Compression • Custom UI Shaders

**Profiling & Debugging**

Unity Profiler • Memory Profiler • Frame Debugger • RenderDoc • Android Logcat / ADB

**Platform & Native Bridges**

Android (JNI / Java) • iOS (Obj-C Bridge) • WebGL (WASM) • OTA Manifest Sync

**Architecture & AI Systems**

Deterministic FSMs • MVP / ScriptableObjects • Zero-GC Object Pooling • Stockfish / A* / Backtracking

**Core Engine & Developer Tooling**

Unity 3D (2020–2023 LTS / Unity 6) • C# (.NET) • HLSL / ShaderLab • Custom EditorWindows • Git / GitLab

## EDUCATION

Academic Qualifications & Computer Science Foundation

**Master of Computer Application (MCA) [ CGPA: 7.52 ]**

**Government Autonomous Engineering College**

Aurangabad, Maharashtra — **Dec 2021 – Jan 2023**

**Bachelor of Computer Science (B.Sc. CS) [ CGPA: 7.50 ]**

**Vivekanand Arts and Science College**

Aurangabad, Maharashtra — **Apr 2018 – Oct 2021**